Chapter 6

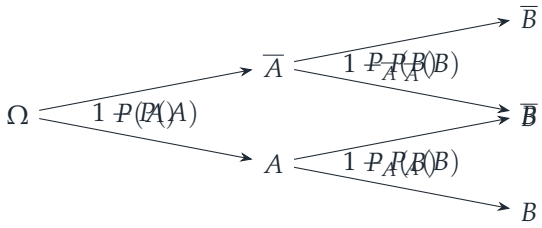
Conditional probability

Skills: model with a tree or a table, calculate, reason about independence, interpret

6.1 Reference course

For two events A and B with $P(A) > 0$, the probability of B given A is

$$P_A(B) = P(B \mid A) = \frac{P(A \cap B)}{P(A)}.$$



Key points

In a weighted tree, the probabilities leaving the same node sum to 1. We multiply the probabilities along a path: $P(A \cap B) = P(A)P_A(B)$. We add the probabilities of mutually exclusive paths leading to the event we are looking for.

If $A_1, \ldots, A_n$ form a partition of the sample space, then

$$P(B) = P(A_1)P_{A_1}(B) + \cdots + P(A_n)P_{A_n}(B).$$

Two events A and B with nonzero probabilities are independent if $P(A \cap B) = P(A)P(B)$, or equivalently if $P_A(B) = P(B)$.

Common mistake

Do not confuse $P_A(B)$ and $P_B(A)$. The condition comes after the word "given" and is written as a subscript in this notation.

Worked example

40 % of customers buy online. Among them, 12 % return their order; among the others, 5 % return it. Then $P(R) = 0.40 \times 0.12 + 0.60 \times 0.05 = 0.078$. The overall return rate is therefore 7.8 %.

Session 39 — Conditioning

Conditional probability

9 Mar 2027 • B: 2 h – C: 1 h

Session objectives

- Define a conditional probability.
- Move from counts to probabilities.
- Interpret.

Fluency – 6 to 8 minutes, no calculator

1. Calculate 20 % of 140. .....
2. What is the sum of the branches leaving a node? .....
3. Calculate 0.7×0.4. .....
4. If $P(A) = 0.37$, calculate $P(\overline{A})$. .....
5. Simplify $\dfrac{18}{30}$. .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: define a conditional probability and move from counts to probabilities.

Investigation

In a survey, 180 customers out of 600 use a premium card and 72 of these 180 customers buy every week. Calculate $P(P)$, $P(P \cap H)$ and $P_P(H)$, then interpret.

Written record

For $P(A) > 0$, $P_A(B) = \dfrac{P(A \cap B)}{P(A)}$. This is the probability of B within the population restricted to A.

Worked example

Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Practice – essential

1. We know $P(A) = 0.35$, $P_A(B) = 0.6$ and $P_{\overline{A}}(B) = 0.8$. Construct the tree.
2. Calculate $P(A \cap B)$ then $P(B)$.
3. Calculate $P_B(A)$ and test independence.

Textbook references

Further reading: Calao, course pp. 132–137; exercises p. 138, nos. 1–5. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 138, exercise 5

A company buys electronic chips from three suppliers F_1, F_2 and F_3: 30 % from F_1, 20 % from F_2. Some chips are faulty: 5 % of the chips from F_1, 3 % of those from F_2 and 10 % of the chips from F_3. A chip is chosen at random. Let the events be F_i: "The chip comes from supplier i" and D: "The chip is faulty".

1. Complete the weighted tree from the textbook.
2. Determine the probability that the chosen chip is faulty.
3. Given that the chip is faulty, determine the probability that it comes from supplier F_3.

Homework

Redo practice exercise 2 with the following data: replace the initial value with 315 and the rate or ratio with 5. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 40 — Weighted trees

Conditional probability

11 Mar 2027 • B: 1 h – C: 2 h

Session objectives

- Construct and complete a tree.
- Read a conditional probability.
- Check the sums.

Fluency – 6 to 8 minutes, no calculator

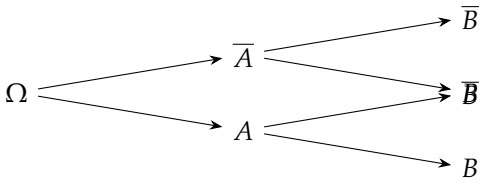
1. Write the multiplier for a 25 % increase. .....
2. Calculate 0.8×0.4. .....
3. If $P(A) = 0.37$, calculate $P(\overline{A})$. .....
4. Simplify $\dfrac{18}{30}$. .....
5. What is the sum of the branches leaving a node? .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: construct and complete a tree and read a conditional probability.

Investigation

35 % of orders are express. Among them, 4 % are late; among standard orders, 9 % are late. Construct and complete the tree.



Written record

In a tree, the branches leaving a node sum to 1. The second-level probabilities are conditional.

Worked example

Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Practice – essential

1. We know $P(A) = 0.40$, $P_A(B) = 0.2$ and $P_{\overline{A}}(B) = 0.4$. Construct the tree.
2. Calculate $P(A \cap B)$ then $P(B)$.

3. Calculate $P_B(A)$ and test independence.

Textbook references

Further reading: Calao, course pp. 133–137; exercises p. 140, nos. 14–18. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 140, exercise 15

1. Complete the tree from the textbook given that $P(A) = \frac{3}{5}$, $P_A(B) = \frac{7}{10}$, $P_A(D) = \frac{1}{5}$, $P_{\overline{A}}(B) = \frac{1}{3}$ and $P_{\overline{A}}(C) = \frac{2}{9}$.
2. Determine $P(A \cap B)$, $P(A \cap D)$ and $P(C)$.

Homework

Redo practice exercise 3 with the following data: replace the initial value with 320 and the rate or ratio with 6. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 41 — Product of probabilities along a path

Conditional probability

16 Mar 2027 • B: 2 h – C: 1 h

Session objectives

- Calculate an intersection.
- Combine paths.
- Translate an event.

Fluency – 6 to 8 minutes, no calculator

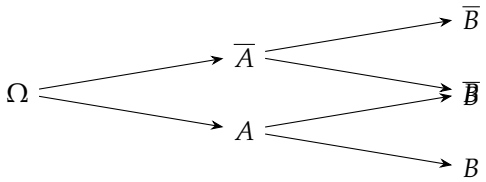
1. Simplify $3x + 9 - x + 2$. .....
2. If $P(A) = 0.37$, calculate $P(\overline{A})$. .....
3. Simplify $\dfrac{18}{30}$. .....
4. What is the sum of the branches leaving a node? .....
5. Calculate 0.9×0.4. .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: calculate an intersection and combine paths.

Investigation

Using the tree from session 40, calculate the probability of an order being express and late, then the total probability of a delay. Interpret both results.



Written record

We multiply along a path: $P(A \cap B) = P(A)P_A(B)$. We add the mutually exclusive paths that realise the event.

Worked example

Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Practice – essential

1. We know $P(A) = 0.45$, $P_A(B) = 0.3$ and $P_{\overline{A}}(B) = 0.5$. Construct the tree.
2. Calculate $P(A \cap B)$ then $P(B)$.

3. Calculate $P_B(A)$ and test independence.

Textbook references

Further reading: Calao, p. 140, nos. 14–18. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 140, exercise 18

75 % of a restaurant's customers are families. Among them, 17 % pay their bill with holiday vouchers, 76 % pay by bank card and the rest pay in cash. Among the other customers, only 5 % use holiday vouchers, 80 % a bank card and the rest cash.
A table and its customer(s) are chosen at random. Let F: "The customers are a family", C: "The customers pay with holiday vouchers", B: "The customers pay by bank card" and L: "The customers pay in cash".

1. Complete the weighted tree from the textbook.
2. Describe the event $\overline{F} \cap B$ in a sentence, then calculate its probability.
3. Calculate the probability that the chosen customers pay with holiday vouchers.
4. The chosen customers paid with holiday vouchers. Calculate the probability that these customers are a family.

Homework

Redo practice exercise 2 with the following data: replace the initial value with 325 and the rate or ratio with 7. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 42 — Independence

Conditional probability

18 Mar 2027 • B: 1 h – C: 2 h

Session objectives

- Test independence.
- Compare probabilities.
- Justify a conclusion.

Fluency – 6 to 8 minutes, no calculator

1. Calculate 5 % of 60. .....
2. Simplify $\frac{18}{30}$. .....
3. What is the sum of the branches leaving a node? .....
4. Calculate 0.3×0.4. .....
5. If $P(A) = 0.37$, calculate $P(\overline{A})$. .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: test independence and compare probabilities.

Investigation

In a population, $P(A) = 0.4$, $P(B) = 0.3$ and $P(A \cap B) = 0.12$. Test independence in two ways. Then change $P(A \cap B)$ to 0.15 and start again.

Written record

A and B are independent if $P(A \cap B) = P(A)P(B)$, equivalently if $P_A(B) = P(B)$ when $P(A) > 0$.

Worked example

Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Practice – essential

1. We know $P(A) = 0.20$, $P_A(B) = 0.4$ and $P_{\overline{A}}(B) = 0.6$. Construct the tree.
2. Calculate $P(A \cap B)$ then $P(B)$.
3. Calculate $P_B(A)$ and test independence.

Textbook references

Further reading: Calao, course pp. 134–137; exercises p. 144, nos. 33–35. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 139, exercise 13

At the exit of a factory, watches may have a defect (denoted D_1) on the strap and a defect (denoted D_2) on the face. A watch is said to be faulty if it has at least one of the two defects. A watch is picked at random from a day's production.
Let A be the event: "The watch has defect D_1" and B the event: "The watch has defect D_2". The probabilities of these events are known: $P(A) = 0.02$ and $P(B) = 0.01$. Assume that these two events are independent.

1. Determine the probability that the watch has both defects.
2. Determine the probability that the watch has no defect.

Homework

Redo practice exercise 3 with the following data: replace the initial value with 330 and the rate or ratio with 2. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 43 — Partition and total probability

Conditional probability

23 Mar 2027 • B: 2 h – C: 1 h

Session objectives

- Recognise a partition.
- Apply total probability.
- Organise the calculation.

Fluency – 6 to 8 minutes, no calculator

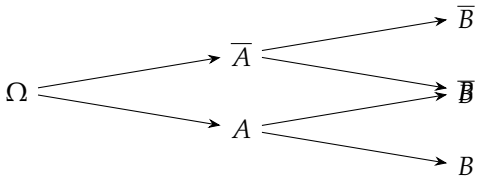
1. Write the multiplier for a 10 % increase. .....
2. What is the sum of the branches leaving a node? .....
3. Calculate 0.4×0.4. .....
4. If $P(A) = 0.37$, calculate $P(\overline{A})$. .....
5. Simplify $\dfrac{18}{30}$. .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: recognise a partition and apply total probability.

Investigation

A factory has three lines A, B, C producing 20 %, 50 % and 30 % of the items respectively. Their defect rates are 1 %, 2 % and 4 %. Calculate the overall defect rate.



Written record

If (A_i) is a partition, $P(B) = \sum_i P(A_i)P_{A_i}(B)$. Each term corresponds to a path leading to B.

Worked example

Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Practice – essential

1. We know $P(A) = 0.25$, $P_A(B) = 0.5$ and $P_{\overline{A}}(B) = 0.7$. Construct the tree.
2. Calculate $P(A \cap B)$ then $P(B)$.
3. Calculate $P_B(A)$ and test independence.

Textbook references

Further reading: Calao, p. 138, nos. 1–5, then p. 142, nos. 22–25. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 139, exercise 7

In the early 2000s, the Internet was not as popular, and a study of interest in the Internet on a sample of 2 000 people gave the following results.

Interested in the Internet?	Yes	No	Total
Under 30	560	···	700
30 to 60	···	···	···
Over 60	···	510	600
Total	800	···	2 000

1. Complete this table.
2. A person is chosen at random from those surveyed. Let A_1: "The person is under 30", A_2: "The person is between 30 and 60", A_3: "The person is over 60" and I: "The person is interested in the Internet". Using the table, complete the probability trees from the textbook.

Homework

Redo practice exercise 2 with the following data: replace the initial value with 335 and the rate or ratio with 3. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 44 — Problems in context

Conditional probability

25 Mar 2027 • B: 1 h – C: 2 h

Session objectives

- Choose a tree or a table.
- Solve a problem in context.
- Interpret.

Fluency – 6 to 8 minutes, no calculator

1. Simplify $3x + 5 - x + 2$. .....
2. Calculate 0.5×0.4. .....
3. If $P(A) = 0.37$, calculate $P(\overline{A})$. .....
4. Simplify $\dfrac{18}{30}$. .....
5. What is the sum of the branches leaving a node? .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: choose a tree or a table and solve a problem in context.

Investigation

Turn a table of counts (customers/subscription/cancellation) into a tree, then answer a question in each direction: probability of cancelling given the subscription; probability of having the subscription given the cancellation.

Written record

Choosing between a tree and a table: the tree highlights the order of conditioning; the table suits counts and reversed readings well.

Worked example

Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Practice – essential

1. We know $P(A) = 0.30$, $P_A(B) = 0.6$ and $P_{\overline{A}}(B) = 0.8$. Construct the tree.
2. Calculate $P(A \cap B)$ then $P(B)$.
3. Calculate $P_B(A)$ and test independence.

Textbook references

Further reading: Calao, p. 142, nos. 22–25, then p. 144, nos. 30–32. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 142, exercise 24

A town hall carries out a survey to learn about its residents' habits regarding waste treatment. 80 % of the people surveyed say they are sensitive to sustainable development, but among them, only 75 % sort their waste. 70 % of the people surveyed say they sort their waste.
One of the people surveyed is chosen at random. Consider the events D: "The person surveyed is sensitive to sustainable development" and T: "The person surveyed sorts their waste".

1. Determine $P_D(T)$.
2. Determine $P_T(D)$.

Homework

Redo practice exercise 3 with the following data: replace the initial value with 340 and the rate or ratio with 4. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 45 — Mixed practice

Conditional probability

30 Mar 2027 • B: 2 h – C: 1 h

Session objectives

- Calculate a reverse probability.
- Chain total probability and conditioning.
- Check the result.

Fluency – 6 to 8 minutes, no calculator

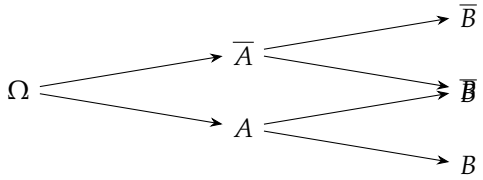
1. Calculate 20 % of 120.
2. If $P(A) = 0.37$, calculate $P(\overline{A})$.
3. Simplify $\dfrac{18}{30}$.
4. What is the sum of the branches leaving a node?
5. Calculate 0.6×0.4.

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: calculate a reverse probability and chain total probability and conditioning.

Investigation

In the situation of session 43, given that an item is defective, calculate the probability that it comes from line C. Explain why this result is not 30 %.



Written record

For a reverse probability, first calculate the total probability of the conditioning event, then use the definition of conditional probability.

Worked example

Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Practice – essential

1. We know $P(A) = 0.35$, $P_A(B) = 0.2$ and $P_{\overline{A}}(B) = 0.4$. Construct the tree.
2. Calculate $P(A \cap B)$ then $P(B)$.

3. Calculate $P_B(A)$ and test independence.

Textbook references

Further reading: Calao, pp. 144–146. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 143, exercise 26

In this exercise, results will be given as irreducible fractions. A game consists of rolling a fair six-sided die and then drawing a ball from an urn. If a 6 is rolled, a ball is drawn from urn A; if an odd number is rolled, the draw is from urn B; in the other cases, the draw is from urn C.
Urn A contains 1 red ball and 5 green balls; urn B contains 2 red balls and 8 green balls; urn C contains 3 red balls and 11 green balls.
Consider the following events: I: "Rolling an odd number"; S: "Rolling 6"; A: "Rolling 2 or 4"; R: "The ball drawn is red"; V: "The ball drawn is green".

1. Draw a weighted tree corresponding to this situation.
2. Determine the probability of drawing a red ball.
3. Determine the probability that the ball comes from urn A given that it is red.

Homework

Redo practice exercise 2 with the following data: replace the initial value with 345 and the rate or ratio with 5. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 46 — Review and assessment

Conditional probability

1 Apr 2027 • B: 1 h – C: 2 h

Session objectives

- Use all the methods.
- Write up with the notation.
- Remedy mistakes.

Fluency – 6 to 8 minutes, no calculator

1. Write the multiplier for a 25 % increase. .....
2. Simplify $\frac{18}{30}$. .....
3. What is the sum of the branches leaving a node? .....
4. Calculate 0.7×0.4. .....
5. If $P(A) = 0.37$, calculate $P(\overline{A})$. .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: use all the methods and write up with the notation.

Investigation

Complete a four-station circuit: complete a tree; calculate an intersection; apply the total probability formula; test independence. Each group leaves a written justification for the next group.

Written record

Always name the events, complete all the branches, write the formula before calculating and interpret the final probability.

Worked example

Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Practice – essential

1. We know $P(A) = 0.40$, $P_A(B) = 0.3$ and $P_{\overline{A}}(B) = 0.5$. Construct the tree.
2. Calculate $P(A \cap B)$ then $P(B)$.
3. Calculate $P_B(A)$ and test independence.

Textbook references

Further reading: Calao, pp. 146–147. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the

textbook.

Calao textbook – p. 143, exercise 29

To hire its managers, a company uses a recruitment agency. The agency first makes a selection based on applications. 40 % of the applications received are accepted and passed on to the company. The selected candidates have a first interview, after which 70 % of them are retained. These are invited to a final interview with the human resources director, who hires 25 % of the candidates met.
A candidate's application is chosen at random. Consider the following events: D: "The candidate is retained on application"; E_1: "The candidate is retained after the first interview"; E_2: "The candidate is hired".

1. Complete the weighted tree from the textbook. Let F be the event: "The candidate is not hired". Prove that the probability of the event F equals 0.93.
2. Five friends apply for a manager's job in this company. Their applications are considered independently of one another. Determine the probability that at least one of the five friends is hired.
3. What is the minimum number of applications the recruitment agency must process for the probability of hiring at least one candidate to exceed 0.999?

Homework

Redo practice exercise 3 with the following data: replace the initial value with 350 and the rate or ratio with 6. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.